Risk Management Plan (RMP)

Project Name: Maternity Stream System (MSS)
Organization: Maternity Stream International Pty Ltd (MSI)

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1 Risk Management Organisation

1.1 Process Responsibility

Risk management is a shared responsibility and involves:

- Developing and implementing Risk Mitigation.
- Generating risk reports, including trends and metric analysis.
- Clarifying, consolidating and documenting risks.
- Maintaining and monitoring data in the Risk Register.
- Monitoring the status of risk mitigation.
- Communicating the status of risks and mitigations to risk owners.
- Escalating communication/action if expected mitigation action deadlines are unlikely to be met.
- Executing the risk closure process.

These responsibilities are undertaken by team members, but the accountability often rests with the Project Lead. Specific Risk Management responsibilities of the Project Lead include:

- Approving the mitigation of very high-severity level risks.
- Supporting mitigation implementation as appropriate.
- Assisting in cross-organisation and controversial risk mitigation to facilitate the involvement of senior personnel from other organisations and their resources.

1.2 Risk Owners

The Risk Owner (RO) is the person responsible for managing/mitigating a specific risk. This assignment of responsibility will be based on the type of risk and will normally be delegated to the team member who can be empowered to ensure this risk is managed/mitigated. This will typically be a Team Leader and/or their respective co-lead. Other stakeholders can also be delegated as Risk Co-Owners (RCO), so appropriate skills, knowledge and experience can be applied to manage and mitigate the risk. The risk owner depends on the type of risk

The RO and RCOs (as appropriate) will take the following actions:

- Monitor the risk throughout its lifecycle.
- Evaluate the impact and likelihood.
- Develop and implement mitigation or contingency plans.
- Communicate updates to the team and stakeholders.
- Escalate if the risk becomes critical or unmanageable.

2 Risk Management Process

Risk management involves five major activities: Identify Risks, Analyse Risks (Qualitative & Quantitative), Plan Risk Responses, Implement Risk Responses, and Monitor Risks throughout the project life cycle. These are discussed in the following section.

2.1 Identify Risks

2.1.1 Create a Risk Breakdown Structure

A Risk Breakdown Structure (RBS) is a hierarchical framework that categorises and organises project risks. It helps teams visualise and classify risks to ensure comprehensive identification and management. Major risk categories (level 0), minor categories (level 1), subcategories (level 2), and specific risks within subcategories (level 3) have been identified.

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Table 1. Risk Breakdown Structure (RBS)

LEVEL 0 (Broad Categories)	**LEVEL 1** (Minor Categories)	**LEVEL 2** (Sub-Categories)	**LEVEL 3** (Specific Risks with Sub-Categories)
Project Risk	1. Project Management and Governance	Stage-gate approvals and change control	**R01:** A delay in approving the execution stage gate on 30 April 2027 could prevent the network installation and production migration activities from starting on time. **R02:** Changes to technical requirements without proper approval could create differences between the agreed project scope and the work being carried out.
Project Risk	Project Management and Governance	Budget and schedule management	**R03:** Higher-than-expected equipment, installation or data migration costs could put pressure on the approved $514,250 budget and its contingency reserves. **R04:** Delays in network commissioning or data reconciliation could affect the planned Go-Live approval on 30 September 2027.
Project Risk	2.Procurement and Contractor Coordination	Equipment and resource availability	**R05:** If MSI cannot provide the required development and testing equipment by January 2027, DemSet may be unable to complete the planned test migration on schedule. **R06:** Late delivery of routers, firewalls, switches or software licenses could delay DCPlus's network installation across Perth, Rockingham and Mandurah.
Project Risk	Procurement and Contractor Coordination	Dependencies between contractors	**R07:** Differences in technical requirements between DCPlus, DemSet and StreamTech could cause problems during system integration and testing. **R08:** Changes to DocCloud's SaaS specifications or integration requirements after the designs have been approved could force contractors to repeat completed work.

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Project Risk	3.Network and Infrastructure	Network design and performance	**R09:** If the initial network assessment misses existing bandwidth limitations, the upgraded infrastructure may struggle to support telehealth video and other MSS services. **R10:** Incorrect IP addressing, VLAN configurations or routing settings could prevent different MSS components from communicating properly.
Project Risk	Network and Infrastructure	Installation, reliability and recovery	**R11:** If redundant network connections do not operate as expected, a network failure could interrupt access to the MSS platform at one or more sites. **R12:** Problems during network cutover could disrupt maternity services if DCPlus cannot restore the previous configuration quickly.
Project Risk	Network and Infrastructure	Existing equipment assessment	**R13 – Opportunity:** If the infrastructure assessment identifies equipment that can safely be reused, MSI may reduce replacement costs and unnecessary electronic waste.
Project Risk	4.Data Migration and Data Quality	Data mapping and cleansing	**R14:** Incorrect mapping between existing patient records and the new MSS data structures could result in missing or inaccurate clinical information. **R15:** Duplicate, incomplete or inconsistent patient records that remain after data cleansing could create errors during final reconciliation.
Project Risk	Data Migration and Data Quality	Production migration and recovery	**R16:** Migration tools that perform well during testing may struggle with the full volume of production data, causing the data transfer to exceed the agreed downtime window. **R17:** A failed production migration could lead to extended service disruption if the data recovery or rollback process does not work as planned.

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Project Risk	Data Migration and Data Quality	Reuse of tested migration processes	**R18 – Opportunity:** Successfully reusing validated migration scripts and cleansing procedures from the test migration could reduce repeated work and improve the production migration process.
Project Risk	5.Security and Privacy	Identity and network security	**R19:** Incorrectly configured user access permissions or multi-factor authentication could allow unauthorized access to sensitive MSS information. **R20:** Firewall or network security settings that are not properly implemented could expose sensitive system connections or prevent authorized access.
Project Risk	Security and Privacy	Protection of patient information	**R21:** Using identifiable patient information in testing without adequate security controls could expose confidential health records. **R22:** Patient data that is transferred or stored without the required security protections could result in a privacy breach during migration.
Project Risk	6.Testing and Quality Management	Testing environments and system integration	**R23:** Delays in preparing the shared testing environment could prevent DemSet, DCPlus and other relevant teams from completing planned testing activities. **R24:** Incomplete performance and integration testing across the three sites could leave network or system problems undiscovered until the MSS platform is in use.
Project Risk	Testing and Quality Management	Quality acceptance and sign-off	**R25:** Serious defects that remain unresolved during the final readiness review could prevent the system from receiving approval for Go-Live. **R26:** If the CIO or required clinical representatives are unavailable to review migration results, formal acceptance of the migrated data could be delayed.
Project Risk	7.Stakeholder Engagement and Operational Readiness	Stakeholder communication and participation	**R27:** Late communication about network upgrades or data migration downtime could leave maternity

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			service staff unprepared for temporary service interruptions. **R28:** Limited participation from clinical users during acceptance testing could allow important problems with everyday workflows to remain unnoticed.
Project Risk	Stakeholder Engagement and Operational Readiness	Training, handover and support	**R29:** Incomplete operational support guides, network documentation or Knowledge Base information could make it difficult for MSI's support team to resolve incidents after implementation. **R30:** Delays in training IT staff and end users could leave employees unprepared to use or support the new system during the transition.

2.1.2 Create a Risk Register

A risk register is a log of possible risks that could affect the project. Prioritised such that high risks are listed first and managed early in the project. Possible risks in each of the RBS categories are analysed in risk register in section 3, FMEA in section 4, and RCA in section 5. Once a risk has been identified, it must be logged into the Project Risk Register. The Risk Register is provided in Section 3.

2.2 Analyse Risks

Risks will be analysed using Quantitative and Qualitative methods. Such analysis can be conducted in parallel or in tandem *(e.g. one after the other)*. Qualitative Analysis is a method for assessing the level of risk by entering the probability of occurrence and the impact of an occurrence using a matrix, such as the one provided in *Figure 1*. This is achieved by defining each issue in terms of the general criteria to the right of this diagram.

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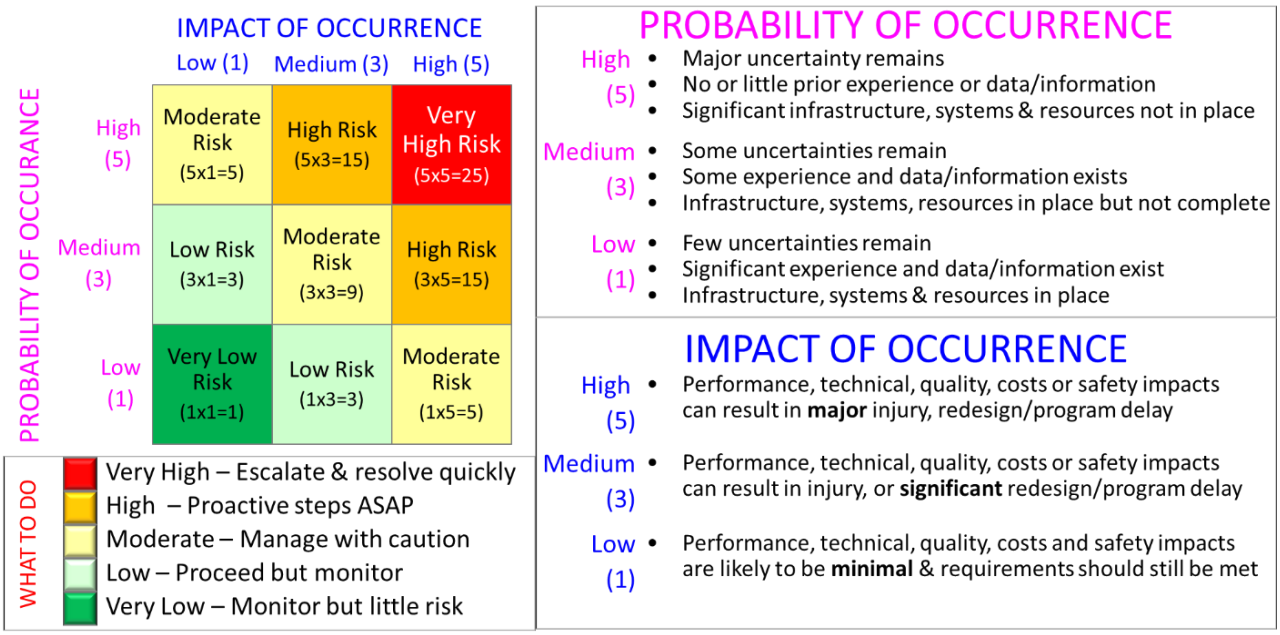


Figure 1: Qualitative Analysis Matrix

Once this qualitative analysis has been completed, take appropriate action as defined in Table 1:

Table 2: Actions to be taken for different levels of Qualitative Risk

Score	Definition/Actions to be taken
Very High (25)	Anything classified as Very High indicates that this risk is extremely or very likely to occur. Additionally, the occurrence could have a profound impact on the project's safety, technical, cost, and/or schedule, which may cause the project to be terminated or can cause significant cost/schedule changes *(e.g. increases of more than 5 percent)*. The management of this level of risk should be escalated, and that aspect of the project must be implemented with *extreme care* until the risks can be mitigated/controlled effectively.
High Risk (15)	High Risks may cause significant safety, technical, cost, and/or schedule increases *(e.g. increases of 2 to 5 percent)* for the project. These risks are to be managed proactively, and a priority must be applied to mitigate/control the risks as soon as practicable. In the meantime, the elements of the project associated with this risk must be managed with due care.
Moderate Risk (5 or 9)	This refers to risks that are Moderate, because they may have a relatively small but significant impact on the project's safety, technical, cost, and/or schedule *(e.g. generally less than 2 percent variance)*. Appropriate mitigation/control strategies should be implemented as soon as it is practicable. Obviously, risks with a score of nine (9), should be addressed with higher priority than those with a score of (5). While awaiting mitigation/controls to be implemented, the team should still manage this aspect of the project with care.
Low Risk (3)	A Low Risk refers to an event that is relatively unlikely to occur, or the impact would be low if it did occur. In other words, this refers to situations in which the combination of likelihood and impact means that this risk would not be expected to have a significant impact on the project's safety, technical, cost and/or schedule. Typically, consolidated risk management is not applied to these types of risks. However, the team associated with this aspect should keep it in mind while implementing the project and monitor the issue with an appropriate level of caution.

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Very Low Risk (1)	A Very Low Risk refers to matters where it would be unlikely for the risk to occur and even if it did, the impact is expected to be minimal. In these circumstances, consolidated risk management would not be applied. However, as with all aspects of Risk Management, those involved with the project should continue to monitor evolving levels of risk and take proactive action when considered appropriate.

2.3 Plan and Implement Risk Responses

Where the risks are identified as being Very High, High, or Moderate, an RO *(and possibly one or more RCO)* will be allocated to manage the risk. They will begin this activity by developing a mitigation strategy (Risk Response Plan) – this provides options and action plans, which can help to reduce the threats associated with the identified risk. Recommended responses, strategies and actions are included and continuously updated in the risk register (section 3), FMEA (section 4), RCA (section 5).

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3 Risk Register

REF ID	Description of the Risk	Potential Impact of the Risk	Risk Level (VH/H/M/L/VL)	Risk Owner/s (RO/RCO)	Date Reported	Control/Contingency/Fallback Strategies	Due Date
1	**RBS R17: Failed migration** The production data migration fails part-way and the rollback or data recovery process does not return the legacy data to its pre-migration state	Extended outage of maternity services at Perth, Rockingham and Mandurah Patient-safety exposure from unavailable records Go-live schedule delay Revision of budget distribution	VH (25)	RO: DemSet Lead RCO: Data Migration Expert, Project Manager	9/10/26	• Control: Take a full verified backup immediately before cutover; write the rollback procedure and rehearse it end-to-end during the test migration; run a formal go/no-go checklist signed off by the Sponsor (CIO) • Contingency: Invoke rollback within the agreed recovery time objective, restoring from the verified backup or DR replica, with the legacy system kept available read-only • Fallback Strategies: Operate on documented manual downtime procedures while the migration is re-planned; Steering Committee approves a new migration window	31/3/2027
2	**RBS R14: Incorrect Data Mapping** Fields in existing patient records are mapped wrongly to the MSS (DocCloud) data structures, so clinical information is missing or inaccurate after migration	Loss of clinical data Revision of migration cycles Delayed reconciliation and sign-off	VH (25)	RO: Data Migration Expert RCO: DemSet Lead, Data Specialists, Clinical Subject Matter Experts (pathology, imaging, pharmacy)	9/10/26	• Control: Produce a field-level mapping specification that is reviewed and signed off by the clinical SMEs and Data Specialists before any migration runs; build automated field-by-field validation checks i.e. record counts, checksums, critical-field comparisons • Contingency: Halt the affected batch, correct the mapping, and re-run only the impacted records; have SMEs verify a sample of corrected records • Fallback Strategies: Clinicians manually verify high-risk records (medications, allergies, obstetric history) against legacy before first use while the legacy system stays accessible	1/12/2026
3	**RBS R19: Misconfigured Access Permission** User roles, permissions or multi-factor authentication in the MSS tenant are set up incorrectly, allowing unauthorised access to confidential maternity records	Loss of patient trust Privacy breach under the Privacy Act Reputational damage Remediation cost	VH (25)	RO: StreamTech Lead RCO: Security and Compliance Consultant, Identity Access Management Team	9/10/26	• Control: Define a role-based access matrix approved by the Security and Compliance Consultant; apply least-privilege roles; enforce MFA for all users; peer-review the configuration and run an independent access-control test before go-live • Contingency: Immediately suspend the affected accounts or roles, review audit logs to establish exposure, and assess notification obligations • Fallback Strategies: Restrict the tenant to an admin-only mode until permissions are rebuilt and re-tested, and engage an independent security assessor to verify	1/2/2027

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4	**RBS R21: Identifiable patient data used in testing** Real patient information is loaded into the shared test environment without adequate security controls	Potential halt of testing Confidential health records exposed to developers and testers who have no clinical need Reputational damage Privacy breach under the Privacy Act	VH (25)	RO: QA and Test Lead RCO: DemSet Lead, Security and Compliance Consultant	9/10/26	• Control: Mandate masked or de-identified data in all non-production environments; complete a privacy impact assessment before the test environment is populated; restrict test environment access by role and log it • Contingency: Stop testing, purge the data, review access logs and determine whether the incident is notifiable • Fallback Strategies: Switch to a fully synthetic data set for the remaining test cycles and obtain an independent privacy review before testing resumes	31/12/2026
5	**RBS R22: Unprotected data transfer or storage during migration** Patient data is moved or stored in transit or staging without the required encryption and access controls	Privacy breach of sensitive patient data Mandatory regulator notification Legal liability for MSI and contractors; migration paused	VH (25)	RO: Security and Compliance Consultant RCO: DemSet Lead, Legal Support	9/10/26	• Control: Require encryption in transit and at rest, Australian data residency, secure transfer channels, and signed data-handling clauses in the DemSet contract; limit staging-area access to named personnel • Contingency: Halt the transfer, contain and delete exposed copies, run the incident response plan and assess notification to the OAIC and affected patients • Fallback Strategies: Move the data using encrypted physical media under chain-of-custody, or an alternative approved secure channel, while the original route is remediated	31/1/2027
6	**RBS R11: Redundant network connections fail to operate** Failover between redundant links does not work when a primary link fails at one or more sites	Loss of access to MSS platform Patient care disruption Breach of availability target	VH (25)	RO: DCPlus Lead RCO: Network Engineers	9/10/26	• Control: Design dual-path connectivity on diverse carriers at every site; test failover at each site during commissioning and again before Go-Live; set up link monitoring and alerts • Contingency: Trigger manual failover and escalate to the carrier under the DCPlus service agreement • Fallback Strategies: Bring up a pre-configured 4G/5G backup connection for critical services and invoke the staff downtime procedures	31/5/2027
7	**RBS R04: Late network commissioning**	Go-live schedule delay Loss of Board Confidence	VH (25)	RO: Project Manager	9/10/26	• Control: Maintain an integrated schedule with critical path and float; hold weekly critical-path reviews with the three	30/11/2026

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	Delays in installing the network or reconciling migrated data push back Go-Live approval, which is a fixed milestone	Pilot benefits and revenue deferred		RCO: DCPlus Lead, DemSet Lead		contractors; define early-warning triggers (e.g. any task more than 5 days late) • Contingency: Fast-track or run tasks in parallel, add contractor resources, and use schedule contingency with Sponsor approval • Fallback Strategies: Request Steering Committee approval for a staged go-live (core services first, remaining sites or features later) rather than moving the approval date	
8	**RBS R16: Migration tools struggle at production volume** Tools that perform well in the test migration cannot move the full production data set within the agreed downtime window	Downtime overruns Services unavailable for longer than planned Added cost Cutover cancelled or repeated	H (15)	RO: DemSet Lead RCO: Data Migration Expert	9/10/26	• Control: Performance-test the migration tools against a production-size, masked copy of the data; use incremental or delta loads so the final cutover moves only recent changes; agree the downtime window with clinical operations • Contingency: Split the migration into site-by-site or record-type batches and use a longer off-peak window agreed in advance • Fallback Strategies: Roll back to legacy and reschedule to the next approved window, keeping legacy live in parallel until reconciliation passes	30/3/2027
9	**RBS R12: Network cutover cannot be reversed quickly** A problem during cutover disrupts services and DCPlus cannot restore the previous configuration in time	Interrupted appointments Patient inconvenience and safety risk Reputational damage	H (15)	RO: DCPlus Lead RCO: Change Manager, Operations Technology Support	9/10/26	• Control: Write a cutover plan with timed steps and rollback triggers; take configuration backups of all devices; rehearse in the test environment; schedule out-of-hours with a go/no-go gate • Contingency: Execute rollback to the last known good configuration within the agreed window • Fallback Strategies: Keep the legacy network live in parallel until the new one is proven, and invoke downtime procedures if needed	25/6/2027
10	**RBS R09: Network assessment misses bandwidth limits** Existing bandwidth constraints at the three sites are not identified, so the upgraded network cannot carry telehealth	Poor video quality or dropped consultations Users abandon the system Rework and additional network cost	H (15)	RO: DCPlus Lead RCO: Network Engineers, Architects	9/10/26	• Control: Measure peak-hour throughput at every site; simulate concurrent video consultations and imaging transfers; design with agreed headroom and traffic prioritisation (QoS) • Contingency: Increase link capacity through DCPlus and apply QoS to prioritise clinical traffic • Fallback Strategies: Degrade (audio-first telehealth, scheduled off-peak imaging transfers) until capacity is increased	31/12/2026

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	video and medical imaging						
11	**RBS R20: Firewall or security settings poorly implemented** Firewall rules are wrong, exposing system connections or blocking authorised users and integrations	Unauthorised access to the network Clinicians locked out of MSS Security and audit failures	H (15)	RO: DCPlus Lead RCO: Security and Compliance Consultant	9/10/26	• Control: Approve a firewall rule baseline reviewed by the Security Consultant; apply all changes through change control; test rules in the test environment first; run a vulnerability scan after installation • Contingency: Roll back to the last known good rule set and fix the faulty rules under emergency change • Fallback Strategies: Switch to a vendor-hardened standard template or standby firewall and commission an external configuration audit	30/4/2027
12	**RBS R15: Duplicate or inconsistent records survive cleansing** Duplicate, incomplete or conflicting patient records remain after data cleansing and cause errors in final reconciliation	Reconciliation fails or is delayed Wrong patient information Clinician rework Sign-off delayed	H (15)	RO: DemSet Lead RCO: Data Specialists, Data Migration Expert	9/10/26	• Control: Run a data quality assessment with agreed thresholds; apply matching rules to find duplicates; repeat cleansing cycles through the test migration until thresholds are met • Contingency: Quarantine failing records and have a small review team resolve them manually • Fallback Strategies: Migrate with exceptions flagged and clear non-critical records after Go-Live, under a documented risk acceptance signed by the CIO and clinical representative	26/2/2027
13	**RBS R24: Incomplete performance and integration testing** Testing across the three sites does not cover load and integrations, so faults are found only in live use	Outages or slow performance after Go-Live Emergency fixes Reputational damage with WA Health	H (15)	RO: QA Test Lead RCO: Test Manager, StreamTech Lead, DCPlus Lead	9/10/26	• Control: Approve a test strategy covering all three sites, peak-load and failover tests, and integrations (EHR, LIS, PACS, billing), with a traceability matrix to requirements • Contingency: Add a targeted test cycle using schedule contingency, prioritising critical clinical scenarios • Fallback Strategies: Go live with a restricted scope plus enhanced hypercare and agreed rollback criteria, approved by the Steering Committee	30/7/2027
14	**RBS R25: Serious defects unresolved at readiness review**	Go-Live approval withheld Deferred benefits	H (15)	RO: QA Test Lead RCO: StreamTech Lead, Product Owner	9/10/26	• Control: Agree defect severity definitions and Go-Live exit criteria up front; hold twice-weekly defect triage with StreamTech from the first test cycle	31/8/2027

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	Severity 1 or 2 defects remain open at the final readiness review, blocking Go-Live approval	Extended hypercare cost Stakeholder confidence lost				• Contingency: Run a dedicated fix-and-retest sprint with daily triage • Fallback Strategies: Seek Steering Committee approval to go live with documented workarounds for lower-severity defects, with fixes scheduled in hypercare	
15	**RBS R05: Test equipment not available by January 2027** MSI cannot supply the development and testing equipment DemSet needs for the test migration on schedule	Test migration delayed Compressing data quality and reconciliation work Knock-on delay to production migration	H (15)	RO: Project Manager RCO: DemSet Lead, DCPlus Lead	9/10/26	• Control: Finalise the bill of materials and place purchase orders by November 2026 after confirming vendor lead times; track delivery weekly • Contingency: Lease or borrow equivalent equipment from DCPlus or the vendor, or use a cloud-hosted test environment • Fallback Strategies: Re-sequence testing to start in the DocCloud sandbox and raise a change request to move the test window using schedule contingency	30/11/2026
16	**RBS R06: Late delivery of routers, firewalls, switches or licences** Hardware or licences arrive late, delaying DCPlus installation across Perth, Rockingham and Mandurah	Installation slips toward the Go-Live date Idle contractor time and extra costs Compressed testing	H (15)	RO: DCPLus Lead RCO: Project Manager	9/10/26	• Control: Order long-lead items early with delivery dates in the contracts; stage deliveries by site; hold weekly supplier status checks • Contingency: Expedite shipping or use DCPlus loan units to install the Perth site first • Fallback Strategies: install interim equivalent equipment and swap in the specified items later under change control	31/1/2027
17	**RBS R07: Contractor technical requirements conflict** DCPlus, DemSet and StreamTech work to different technical requirements, causing integration failures	Integration and test failures Rework Blame between contractors Delay and cost increase	H (15)	RO: Project Manager RCO: Architects, DemSet Lead, DCPlus Lead, StreamTech Lead	9/10/26	• Control: Maintain one integrated technical requirements baseline and interface control document; hold joint design reviews before each stage gate • Contingency: Convene a resolution workshop with the Architects and decide through change control • Fallback Strategies: Escalate through contract clauses and bring in an independent integration specialist	31/12/2026

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18	**RBS R01: Execution stage gate approved late** Approval of the Execute stage gate is delayed, so network installation and production migration cannot start on time	Installation and migration start late, squeezing the remaining schedule to Go-Live Contractors on standby	H (15)	RO: Project Manager RCO: Project Sponsor, Steering Committee	9/10/26	• Control: Use a stage-gate readiness checklist and circulate papers two weeks ahead; book Steering Committee dates early • Contingency: Request an out-of-cycle decision meeting or circulate a decision paper • Fallback Strategies: Seek conditional approval for long-lead activities and use delegated Sponsor authority while the full gate is completed	30/4/2027
19	**RBS R03: Cost pressure on the approved budget** Equipment, installation or migration costs exceed estimates and consume the contingency reserve	Contingency exhausted Scope cuts or additional funding request Loan servicing burden rises	H (15)	RO: Project Manager RCO: Project Sponsor	9/10/26	• Control: Track cost against budget monthly using earned value; use fixed-price contracts where possible; obtain firm quotes before ordering • Contingency: Draw on the contingency reserve with Sponsor approval and de-scope lower-priority customisations • Fallback Strategies: Request additional funding from the Steering Committee or move purchases to leasing	31/1/2027
20	**RBS RS10: Incorrect IP, VLAN or routing settings** Network addressing or routing errors stop MSS components and sites communicating	Integrations and sites fail to connect Testing blocked Troubleshooting delays	H (15)	RO: DCPlus Lead RCO: Network Engineers, Infrastructure Engineer	9/10/26	• Control: Document an IP and VLAN plan reviewed by the Architects; validate it in the test environment before installation • Contingency: Apply the validated configuration template and correct faulty settings under emergency change • Fallback Strategies: Revert to the documented baseline and have DCPlus senior engineers rebuild the affected segment	30/3/2027
21	**RBS R23: Shared test environment not ready** The shared test environment is not ready, so DemSet, DCPlus and StreamTech cannot complete planned test activities	Testing delayed Reduced test coverage Go-Live readiness at risk	H (15)	RO: QA and Test Lead RCO: Project Manager, StreamTech Lead	9/10/26	• Control: Set an environment readiness date with entry criteria and a booking schedule across contractors • Contingency: Prioritise critical test activities and add environment build resources • Fallback Strategies: Use the DocCloud sandbox temporarily for testing and adjust the test plan through change control	31/12/2026
22	**RBS R26: Acceptance reviewers unavailable**	Reconciliation sign-off delayed	H (15)	RO: Project Manager	9/10/26	• Control: Book review sessions and named delegates well ahead; issue reconciliation reports in a standard format	30/7/2027

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	The clinical representatives cannot review migration results, delaying acceptance	Go-Live approval at risk		RCO: Project Sponsor, Clinical Subject Matter Expert		• Contingency: Appoint a delegated approver with Sponsor authority • Fallback Strategies: Hold an extra review session and escalate to the Steering Committee	
23	**RBS R27: Late communication of downtime** Maternity staff are not told in time about network upgrades or migration downtime and are unprepared	Disrupted appointments Patient safety risk Complaints	H (15)	RO: Change Manager RCO: Project Manager, Communications Specialist	9/10/26	• Control: Publish a communications plan with notice periods; confirm receipt by team leaders • Contingency: Send targeted urgent notices and brief team leaders in person • Fallback Strategies: Reschedule the downtime if staff readiness cannot be confirmed	30/7/27
24	**RBS R29: Incomplete support documentation** Operational support guides, network documentation or Knowledge Base content is incomplete at handover	Slower incident resolution Longer hypercare Weak Level 1 to 3 support	H (15)	RO: Help Desk Manager RCO: Project Manager, StreamTech Lead, DCPlus Lead	9/10/26	• Control: Set documentation templates and deadlines per contractor; review as-built documentation before the go-live checklist • Contingency: Assign MSI staff to complete gaps with contractor support • Fallback Strategies: Extend contractor warranty support until documentation is complete	31/8/2027
25	**RBS R30: Delayed training of IT staff and users** Training is late, leaving staff unprepared for the new system	Low adoption Support tickets increase Clinical workflow errors	H (15)	RO: Change Manager RCO: Project Manager, Trainer, Help Desk Manager	9/10/26	• Control: Confirm training dates early and agree release of staff with managers; prepare training material ahead of Go-Live • Contingency: Run additional sessions and short refresher videos • Fallback Strategies: Place super-users on each site for the first weeks after Go-Live	30/7/2027
26	**RBS R02: Unapproved changes to technical requirements**	Scope creep and rework Cost pressure on the budget	H (15)	RO: Project Manager RCO: Architect, StreamTech Lead, Project Sponsor	9/10/26	• Control: Baseline the requirements and place them under formal change control (change request, impact assessment, Steering Committee approval above an agreed cost or schedule threshold); keep one requirements register visible to all three contractors	31/12/2026

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REF ID	Description of the Risk	Potential Impact of the Risk	Risk Level (VH/H/M/L/VL)	Risk Owner/s (RO/RCO)	Date Reported	Control/Contingency/Fallback Strategies	Due Date
	Technical requirements are changed without going through change control, so the work delivered drifts from the approved scope	Contractors building to different versions of the requirements Disputes over what is in or out of contract				• Contingency: Pause any unapproved work, assess its cost and schedule impact, then either approve it through change control or revert to the baseline • Fallback Strategies: Escalate under the contract variation clauses, with Legal Support reviewing any contractor claims	
27	**RBS R08: DocCloud changes SaaS specifications after design approval** DocCloud alters its SaaS specification, APIs or integration requirements after designs are approved, forcing StreamTech, DemSet and DCPlus to repeat completed work	Rework and extra cost Integration testing delayed Configurations that no longer work with the vendor release Go-Live phase put at risk	H (15)	RO: StreamTech Lead RCO: Project Manager, Legal Support, Architect	9/10/26	• Control: Write change-notification terms into the DocCloud contract (minimum notice, version freeze and release calendar around Go-Live); favour configuration over customisation; review DocCloud's release roadmap monthly • Contingency: Run an impact assessment on the change, schedule the rework in the next sprint using contingency, and stay on the existing release where DocCloud allows • Fallback Strategies: Invoke contract remedies and negotiate a vendor-supported workaround or credit, led by Legal Support	31/1/2027

4 Failure Mode and Effects Analysis (FMEA) – Priority Risks

This FMEA covers very high risks or associated risks identified in the risk register. The top 10 risks are analysed below as failure modes of MSS components and ranked by Risk Priority Number (RPN). Ratings and assumptions are explained below the table.

Table 3. FMEA Analysis for the MSS

MSS Component / Function[5]	Potential Failure Mode	Potential Failure Effect	Severity Rating[1]	Potential Causes	Occurrence Rating[2]	Detectability Rating[3]	RPN[4]	Recommended Action

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Data Migration – Field Mapping and Transformation (R14)	Legacy fields are mapped to the wrong DocCloud fields or dropped during transformation.	Clinicians see missing or wrong allergies, medications or obstetric history and may make unsafe care decisions.	9	Legacy record formats differ across Perth, Rockingham and Mandurah. Mapping specification is not clinically validated.	6	6	**324**	• Clinical SMEs sign off a field-level mapping specification before any migration run. • Automated field-by-field checks on critical fields, plus a clinician sample review each test cycle.
Data Migration – Production Cutover and Rollback (R17)	Production migration fails part-way and rollback cannot restore the legacy data.	Clinicians at all three sites lose access to patient records. Manual downtime procedures. Go-Live delayed.	10	Rollback never rehearsed at full volume. Backup not verified before cutover.	4	7	**280**	• Rehearse rollback end-to-end during the test migration and time it. • Verified backup and Sponsor-signed go/no-go checklist before cutover
Security Gateway / IAM – User Roles and MFA (R19)	Roles grant wider access than needed, or MFA is not enforced for some accounts.	Staff without a clinical need can view confidential maternity records. Privacy Act breaches.	9	Role matrix not finalised before StreamTech configures the tenant. Errors mapping the Active Directory groups to MSS roles.	5	6	**270**	• Approved role-based access matrix and MFA enforced by policy for all users. • Independent access-control test before Go-Live with added audit-log alerts for unusual access.
Network – Redundant Site Links (R11)	Failover to the secondary link does not trigger when the primary link fails.	Site loses access to the SaaS platform. Telehealth calls drop. Patient care disrupted.	9	Both links share a physical path or carrier. Failover never tested under load.	4	6	**216**	• Diverse carriers and paths at every site. • Failover test at each site during commissioning and before Go-Live, needs to be linked to monitoring and alerts.

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Network – Bandwidth for Telehealth and Imaging (R09)	Site links congest at peak times.	Video consultations freeze or drop. Imaging slow to load. Clinicians revert to old systems.	7	Assessment based on current average use. New video and imaging load underestimated. No traffic prioritisation.	6	5	**210**	• Measure peak-hour use and simulate concurrent video and imaging traffic. • Design with agreed headroom and QoS that prioritises clinical traffic.
Data Migration – Production Data Load (R16)	Bulk load runs past the agreed downtime window.	MSS and legacy systems both unavailable for longer than planned. Appointments disrupted.	7	Tools tested only on a data subset. DocCloud API throttling at full volume.	6	4	**168**	• Full-volume dry run using a masked production copy. • Incremental (delta) loads so cutover moves only recent changes.
Test Environment – Test Data (R21)	Identifiable patient data is loaded into the shared test environment.	Records exposed to contractor staff with no clinical need. Notifiable privacy breach.	8	'Representative data' met by copying live data under time pressure. No masking standard.	5	4	**160**	• Mandatory masking or synthetic data in all non-production environments. • Privacy impact assessment before loading and role-based, logged access.
Network – Cutover (R12)	A fault occurs during cutover and the previous configuration cannot be restored quickly.	Sites offline. Appointments and telehealth sessions interrupted.	8	No device configuration backups. Cutover steps not rehearsed.	4	5	**160**	• Timed cutover plan with rollback triggers and configuration backups. • Rehearse in the test environment, specifically in the out-of-hours window with go/no-go gate.

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Network Security – Firewall (R20)	Rules are too open, or block service bus and API gateway traffic to EHR, LIS and PACS.	Exposure to intrusion, or clinicians and integrations blocked.	8	Rules built ad hoc and changed outside change control.	5	4	**160**	• Security Consultant-approved rule baseline, changed only through change control. • Test rules in the test environment. • Vulnerability scans after installation.
Data Bus – Migration Transfer and Staging (R22)	Patient data is moved or staged without encryption.	Privacy breaches. OAIC notifications. Migration paused.	10	Ad-hoc transfer methods. Staging area set up without a security review.	3	5	**150**	• Encryption in transit and at rest, and Australian data residency, written into the DemSet contract. • Security review of the staging area with access limited to named staff.

[1]Severity of the impact of failure (SEV) is scored on a scale of 1 to 10, with the high score assigned to potentially high-impact events

[2]Likely Frequency of occurrence of failure (OCC) event is scored on a scale of 1 to 10, with a high score assigned to events that are likely to occur frequently

[3]The ability of known control methods to detect failure before it causes harm (DET). This is scored on a scale of 1 to 10. A failure mode that could be easily detected by known control methods should be assigned a low score

[4]Risk Priority Number (RPN) is calculated as Severity × Occurrence × Detectability

[5]Remember that a single functionality could face 1 or more potential failures. Each potential failure must be presented in a separate row.

Explanation of ratings and assumptions

- **Scope: T**he ten highest-priority register risks that are failures of an MSS component. R04 (late network commissioning) is a schedule risk rather than a component failure, so it is covered in the RCA (Risk 4) and R20 takes its place.
- **Pre-mitigation ratings: S**cores assume the PMP baseline before the recommended actions are applied. They will be re-scored once actions are in place and reviewed at each stage gate.
- **Severity (S):** 9–10 where a failure could harm patient safety or breach the Privacy Act (wrong clinical data, failed rollback, unauthorised access, unprotected transfer). 7–8 where service is disrupted or degraded but workarounds exist (downtime procedures, audio-first telehealth).
- **Occurrence (O):** 6 where conditions on this project make the cause likely – three sites with different legacy formats (R14), testing on a data subset (R16), and new video and imaging load not in current usage (R09). 3 for R22, as encryption and Australian data residency are standard SaaS and contract terms.

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- **Detectability (D):** 6–7 where the failure stays hidden until it is live or invoked – a rollback is only proven when needed, over-broad roles are invisible to users, and failover is only exercised when a link fails. 4 where testing or scanning reliably catches it (dry-run timings, privacy impact assessment, vulnerability scans).
- **Ranking:** RPN re-orders the qualitative register because it weighs detectability – R14 ranks first (324) as mapping errors are both likely and hard to spot. R22 has the lowest RPN (150) but keeps its Very High register rating; any failure mode with Severity 9 or above is actioned regardless of RPN.

5 Root Cause Analysis (RCA) – Priority Risks

This table highlights potential root causes of the top project risks. Potential root causes are identified, and Recommended Actions are provided. Four project-wide problems were selected by grouping related high-priority risks from the risk register (Section 3) and FMEA (Section 4), and each was analysed using the 5 Whys approach.

Table 4: RCA for Highest Priority Risks

5 Whys		**Potential Risk 1**	**Potential Risk 2**	**Potential Risk 3**	**Potential Risk 4**
Define the potential problem		**Migrated patient records are incomplete or inaccurate when the MSS goes live (R14, R15, R17)**	**Patient information is accessed or exposed without authorisation during the project (R19, R21, R22)**	**The upgraded network cannot deliver the availability and performance the MSS needs at Perth, Rockingham and Mandurah (R09, R11, R12)**	**Go-Live approval on 30 September 2027 is missed because of delays between MSI and the three contractors (R04, R05, R06, R07)**
(a)	Why would this problem happen?	If legacy fields such as allergies, medications and obstetric history are mapped or cleansed incorrectly, and the errors pass reconciliation unnoticed.	User roles, test data and transfer channels are set up differently by three contractors, leaving gaps such as broad roles, unmasked test data or unencrypted staging.	Failover, bandwidth limit or cutover rollback do not work as designed under real clinical load.	Network commissioning, testing or data reconciliation finish late and the readiness review cannot approve Go-Live.
(b)	Why would (a) happen?	The mapping specification and reconciliation checks are written by DemSet's technical team and focus on record counts and checksums rather than clinical meaning.	Each contractor works from its own security assumptions – DemSet's security protocols may not be final when StreamTech configures the tenant and populates the test environment.	They are designed from an assessment of current average usage and tested component by component, not end-to-end at peak (concurrent video and imaging).	Hand-offs between MSI procurement, DCPlus, DemSet and StreamTech slip and the delay cascades along the critical path (e.g. late test equipment delays the test migration).
(c)	Why would (b) happen?	Clinical SMEs (midwifery, pathology, imaging and	The schedule runs tenant configuration and test	DCPlus rated network risk as low (existing fibre redundancy and	Each contractor tracks its own schedule, so slippage on hand-offs

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		pharmacy) are not scheduled to review the mapping rules or sample migrated records.	environment set-up in parallel with security protocol development to protect the Go-Live date.	telco agreements), so less assessment and testing effort was planned.	is not visible to the Project Manager until it affects the critical path.
(d)	Why would (c) happen?	Clinicians' time on project work was not planned or released by site managers, as data migration is treated as a purely technical task.	The security plan is treated as a contractor deliverable rather than a project-wide entry condition controlled by MSI.	MSI accepted the contractor's self-assessment without setting its own measurable network requirements per site.	There are no single integrated master schedules or hand-off agreements with shared dates and acceptance criteria.
(e)	Why would (d) happen?	No one in MSI is made accountable for the clinical quality of the migrated data - only for delivering the migration on time.	MSI has not defined minimum privacy and security criteria that must be met before any contractor handles patient data.	There were no clinically based non-functional requirements (availability, latency, bandwidth) written into the DCPlus contract – success is defined as 'installed' rather than 'performs'.	The contracts were set separately and MSI has not set up integration governance across the contractors.
Root Cause:		**No MSI business owner is accountable for clinical data quality, so migration acceptance is defined by technical measures, and clinical validation is not resourced.**	**Security and privacy controls are not set as mandatory, MSI-owned gate criteria, so contractors can start handling patient data before a common security baseline exists.**	**Network acceptance is not tied to measurable, clinically based non-functional requirements, so the project relies on the contractor's own low-risk assessment.**	**MSI lacks integrated schedule and hand-off governance across contractors, so dependency slippage is discovered too late to recover.**
Recommended Action(s): (If your proposed answer here is something that the company might not have control over, such as external variables, go back up to the previous why-level). The aim is to develop solutions that can support the successful delivery of the project)		• Appoint a Clinical Data Owner (senior midwife) as Risk Co-Owner for R14, R15 and R17, with authority to sign off the mapping specification and reconciliation results. • Add clinical acceptance criteria (100% match on critical fields such as allergies, medications and estimated due date) to the migration exit criteria and go/no-go checklist. • Agree release of clinical SME time with site managers in the resource plan, booked around the January 2027 test migration and the production migration.	• Make an approved Security Baseline (role-based access, data masking standard, encryption and Australian data residency) an entry criterion for tenant configuration, test data loading and any data transfer. • Security and Compliance Consultant completes a privacy impact assessment and signs off each gate with Legal Support adding data-handling clauses to all three contracts. • Run an independent access-control test before Go-Live and report security status in each Sponsor status report.	• Define non-functional requirements per site (e.g. 99.9% availability, peak concurrent telehealth and imaging bandwidth, failover time) with the Telehealth Specialist and clinicians, and add them to the DCPlus statement of work as acceptance criteria. • Require MSI-witnessed end-to-end peak-load and failover tests at each site before the execution stage gate and again before Go-Live. • Make a rehearsed cutover and documented rollback a go/no-go condition for each site.	• Baseline one integrated master schedule covering MSI procurement and all contractor milestones with contractors reporting weekly. • Write hand-off agreements (date, deliverable, acceptance criteria) between contractors into the contracts, and order long-lead equipment by November 2026. • Hold a weekly cross-contractor coordination meeting with critical-path review and early-warning triggers (any task more than 5 days late), escalating to the Sponsor and Steering

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				Committee as per the Communication Plan.

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